

CDS-492: Capstone for Data Science

Fall 2025

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Research Question: How do different weather conditions affect wireless internet performance in Northern Virginia, and is weather actually a significant predictor of connectivity issues?

Rain, temperature extremes, and high winds are commonly blamed for slow internet speeds in rural areas. The assumption is that atmospheric interference degrades wireless signals. But does weather actually matter as much as we think?

This study tests whether meteorological conditions meaningfully predict internet performance, or if we've been blaming the weather for problems that stem from network infrastructure and usage patterns.

Significance

This research challenges the assumption that weather is a major factor in rural internet performance. By showing that network infrastructure and usage patterns matter more than meteorological conditions, this study redirects policy focus toward the actual problems; infrastructure upgrades and network capacity rather than weather-proofing systems. Understanding what really drives connectivity issues has real implications for rural communities that depend on reliable internet access for telemedicine, remote work, and emergency communications.

Prior research

Luomala and Hakala (2015) documented weather effects on signal strength in controlled laboratory settings using 2.4 GHz wireless systems.

Kale (2021) found a correlation of -0.36 between temperature and WiFi transmission failures, explaining only 13% of variance.

These studies focus on theoretical models and controlled experiments rather than real-world internet performance, leaving practical significance of weather effects versus infrastructure factors untested.

Deliverables:

- Capstone project paper,
- Science paper, and
- Slide presentation.

Method of Inquiry Part 1

Weather data collected from Washington Dulles International Airport (KIAD) weather station using Automated Surface Observing Systems (ASOS), providing hourly measurements of temperature, precipitation, humidity, wind speed, and visibility from August 2022 to August 2025.



Method of Inquiry Part 2

Internet performance data collected from Measurement Lab's database via Google Cloud BigQuery. Average download speeds and test counts gathered hourly from 7 Northern Virginia cities: Leesburg, Manassas, Centreville, Sterling, Herndon, Gaithersburg, and Rockville. Data spans the same 3-year period, enabling direct comparison with weather conditions.



Method of Data Collection

The final dataset combined weather and internet measurements into over 20,000 hourly observations including these features:

<i>Column</i>	<i>Unit</i>	<i>Comment</i>
Date	Date	the date at observation
Hour	Hour	the hour at observation
Temperature	C°	the temperature in Celsius
Humidity	%	the percentage of humidity
Precipitation	mm	the amount of precipitation in mm
Visibility	miles	the furthest distance visible in miles
Wind Speed	kph	the wind speed in kph
Download Speed	Mbps	the download speed in Mbps

Theory

Collect M-Lab internet speed data and NOAA weather data for Northern Virginia, aligning timestamps to match weather conditions with performance measurements from August 2022 to August 2025.

Test correlations between weather variables (temperature, precipitation, humidity, wind, visibility) and download speeds to identify relationships.

Build three regression models, weather-only, time-based (hour), and combined, to compare explanatory power using R^2 and F-tests.

Analyze extreme weather conditions (90th percentile precipitation, wind, temperature) using t-tests and ANOVA to test if even severe weather shows practically significant effects. Distinguish between statistical significance and practical significance using effect size measures.

The project integrates what I learned in:

STAT-350: Introductory Statistics II,

CDS-301 Scientific Information and Data Visualization, and

CDS-303 Data Mining

First Hypothesis

H_1 : Weather conditions and internet performance

H_0 : Weather conditions do not significantly predict internet performance

H_a : Weather conditions significantly predict internet performance

Results

Weather-only model: $R^2 = 0.009$ (0.9% variance explained)

Reject H_0 (statistically significant, $p < 0.001$)

However, the effect is practically meaningless; a temperature change of 10°C affects speed by only 6.8 Mbps

Residuals:

Min	1Q	Median	3Q	Max
-174.38	-52.27	-5.85	38.12	487.02

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	200.233464	4.755227	42.108	< 2e-16 ***
Temperature	-0.679456	0.053769	-12.637	< 2e-16 ***
Relative.Humidity	-0.042996	0.030891	-1.392	0.1640
One.Hour.Precipitation	-0.013580	0.007029	-1.932	0.0534 .
Wind.Speed	0.319486	0.069099	4.624	3.79e-06 ***
Visibility	-0.542457	0.349399	-1.553	0.1205

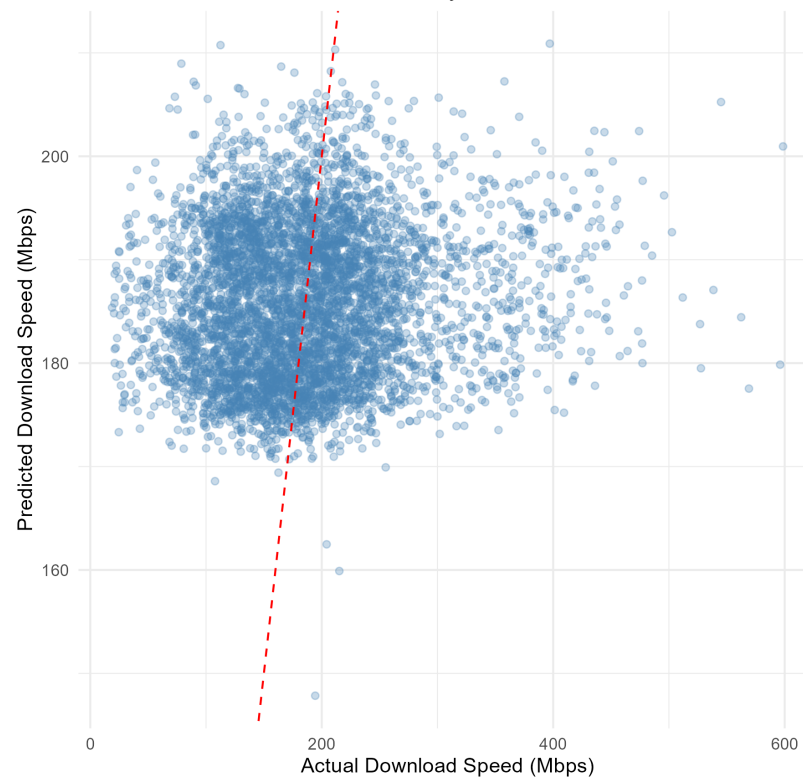
Signif. codes: 0 '***' 0.001 '**' 0.01 '.' 0.05 ' ' 1

Residual standard error: 76.35 on 20970 degrees of freedom

Multiple R-squared: 0.009252, Adjusted R-squared: 0.009015

F-statistic: 39.16 on 5 and 20970 DF, p-value: < 2.2e-16

Actual vs Predicted: Weather-Only Model



Second Hypothesis

H_2 : Time-based factors vs weather as predictors

H_0 : Time-based factors (hour of day) do not explain more variance in internet performance than weather conditions

H_a : Time-based factors explain more variance than weather conditions

Results

Weather model: $R^2 = 0.009$ (0.9%)

Time model: $R^2 = 0.005$ (0.5%)

Fail to reject H_0 (time actually explained LESS variance than weather)

Both models are essentially useless for prediction

Residuals:

Min	1Q	Median	3Q	Max
-174.15	-51.79	-5.94	38.98	485.49

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	194.90590	1.02295	190.53	<2e-16 ***
Hour	-0.80053	0.07631	-10.49	<2e-16 ***

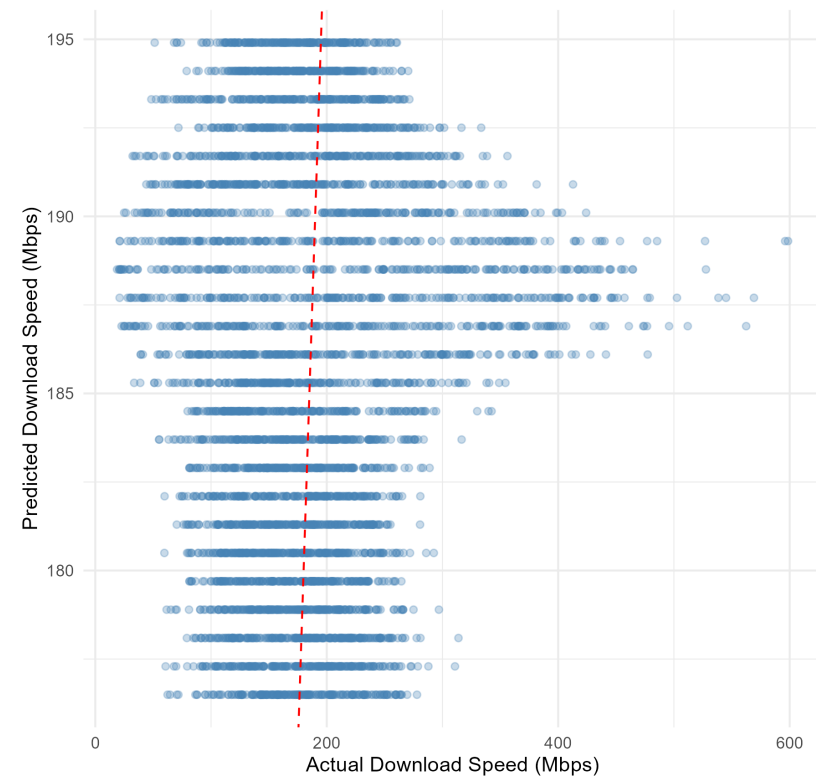
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Residual standard error: 76.5 on 20974 degrees of freedom

Multiple R-squared: 0.00522, Adjusted R-squared: 0.005172

F-statistic: 110.1 on 1 and 20974 DF, p-value: < 2.2e-16

Actual vs Predicted: Time-Based Model



Third Hypothesis

H_3 : Extreme weather effects

H_0 : Extreme weather conditions do not have larger effects on internet performance than typical conditions

H_a : Extreme weather conditions have significantly larger effects than typical conditions

Results

Heavy rain vs normal: 0.033% variance explained, $p = 0.43$

High wind vs normal: 1% variance explained, $p < 0.001$, Cohen's $d = 0.113$

Extreme temperature vs normal: 0.45% variance (largest effect: 22.8 Mbps difference), Cohen's $d = 0.12$

Reject H_0 (statistically significant)

However, even extreme conditions showed negligible practical effects

Source Materials

For Luomala and Hakala (2015), see

<https://annals-csis.org/proceedings/2015/pliks/241.pdf>

For Kale (2021), see

<https://ijrpr.com/uploads/V2ISSUE11/IJRPR1732.pdf>

For internet performance data from Virginia, see

<https://measurementlab.net>

For weather data from IAD Weather Station, see

<https://www.weather.gov/wrh/timeseries?site=KIAD>.